
1st NeurIPS Workshop Proposal on The Data Revolution Behind Modern AI Systems

Website: <https://drbai-2026.github.io/>

Tagline: Data as a first-class foundation: designing, auditing, and governing the fuel of modern AI.

1 Workshop Topic and Content

Modern AI progress is increasingly driven not only by model architecture and compute, but also by the scale, quality, diversity, curation, governance, and transformation of data. Recent breakthroughs in large language models, multimodal foundation models, diffusion models, embodied agents, scientific AI, and autonomous systems all depend on complex data pipelines that span collection, filtering, annotation, synthesis, compression, alignment, evaluation, and continual refinement. However, the research community still lacks a unified forum for systematically studying the *data revolution* behind modern AI: what data is needed, how it should be constructed, how it shapes model behavior, and how it can be made more efficient, trustworthy, and socially beneficial.

This workshop aims to bring together researchers and practitioners working on data-centric AI, foundation model training, dataset construction, synthetic data, data quality evaluation, data attribution, data governance, and AI safety. The central goal is to understand how data acts as a primary driver of modern AI capability, generalization, robustness, reasoning, alignment, and failure modes. Instead of treating data as a static input to model training, the workshop emphasizes data as an active research object: a resource to be designed, optimized, audited, compressed, synthesized, protected, and governed. This workshop provides a timely venue for consolidating emerging efforts in data-centric AI and for shaping a research agenda in which data is treated as a first-class foundation.

The workshop will cover both theoretical and practical aspects of modern AI system data pipelines. Topics include large-scale pretraining corpora, multimodal datasets, instruction and preference data, synthetic data generation, data mixture optimization, dataset distillation, data valuation, benchmark design, data contamination, privacy-preserving data use, and human-AI data production workflows. It will also examine how data affects downstream behaviors such as hallucination, bias, memorization, factuality, safety, robustness, multilinguality, and domain adaptation.

The workshop will feature invited talks, contributed papers, panel discussions, and poster sessions. We aim to boost interdisciplinary discussion among machine learning researchers, data engineers, domain experts, legal and policy scholars, and practitioners building real-world AI systems. By connecting these communities, the workshop seeks to identify open challenges, establish shared evaluation standards, and inspire new research directions for data-centric foundations of modern AI.

Overview of Discussion Topics. We organize topics along the data lifecycle (*construct* → *optimize* → *evaluate* → *ensure trust*), together with the *modalities*, *domains*, and *temporal dynamics* that cut across every stage. The workshop will cover topics including but not limited to:

(1) Data Construction and Synthesis: where training data comes from and how it is built.

- Data collection, filtering, deduplication, and curation for foundation models;
- Synthetic data generation, verification, and evaluation;
- Instruction tuning, preference data, human feedback, and alignment datasets;
- Human-AI collaborative data creation, annotation, and revision workflows.

(2) Data Optimization and Efficiency: which data, and in what proportion, makes modern AI more efficient under a limited budget.

- Data mixture optimization and scaling laws for pretraining, post-training, and domain adaptation;
- Dataset distillation, data compression, data pruning, and compute-efficient data selection.

(3) Data Evaluation, Attribution, and Quality: measuring what a dataset contains and tracing how it shapes model behavior.

- Data attribution, data valuation, influence estimation, and memorization analysis;
- Data contamination, benchmark leakage, and evaluation reliability;
- Data quality metrics, diversity measures, coverage analysis, and distribution-shift detection.

(4) Data Behavior, Safety, and Governance: a primary driver of model behavior and failure modes whose use must be trustworthy and socially beneficial.

- Data-centric approaches to hallucination reduction, factuality, robustness, and safety;
- Privacy, copyright, licensing, governance, and responsible use of large-scale datasets.

(5) Modalities, Domains, and Temporal Dynamics: how these questions change across modalities and domains, and through the continual refinement of data over time.

- Multimodal datasets for language, vision, audio, video, robotics, agents, and scientific AI;
- Continual data updating, dynamic benchmarks, and lifelong data management;
- Domain-specific data pipelines for healthcare, science, education, agriculture, finance, climate, and other high-impact applications.
- Empirical studies, theoretical analyses, benchmarks, systems, tools, and best practices for data-driven AI development.

2 Tentative Workshop Schedule

We have designed a *discussion-centric* schedule: a substantial share of the event is allocated to open discussions. Our hybrid workshop will feature 8 invited talks (30 min each including QAs), 3 oral presentations (10 min each), two 1-hour poster sessions, and a 1-hour panel discussion. All sessions will be accessible for in-person and virtual attendance.

Morning session:

- 09:00 - 09:10 Opening Remarks
- 09:10 - 09:40 Invited Talk 1 (Chelsea Finn)
- 09:40 - 10:10 Invited Talk 2 (Jiaming Song)
- 10:10 - 11:10 Poster Session 1 & coffee socials
- 11:10 - 11:40 Invited Talk 3 (Danqi Chen)
- 11:40 - 12:10 Invited Talk 4 (Zico Kolter)
- 12:10 - 12:20 Oral Presentation 1
- 12:20 - 12:30 Oral Presentation 2
- 12:30 - 13:30 Lunch Break

Afternoon session:

- 13:30 - 14:00 Invited Talk 5 (Zhuang Liu)
- 14:00 - 14:30 Invited Talk 6 (Arash Vahdat)
- 14:30 - 15:30 Poster Session 2 & coffee socials
- 15:30 - 16:00 Invited Talk 7 (Stefano Ermon)
- 16:00 - 16:30 Invited Talk 8 (Bowen Cheng)
- 16:30 - 16:40 Oral Presentation 3
- 16:40 - 17:40 Panel Discussion
- 17:40 - 18:00 Paper Awards and Closing Remarks

3 Invited Speakers

We have invited nine distinguished researchers in the field from both academia, including Stanford, Princeton, and CMU, and industry, including Luma AI, NVIDIA, and Meta. The first seven speakers have already confirmed their participation, while the remaining two are still finalizing their schedules.

Chelsea Finn (Stanford) is an Assistant Professor at Stanford University working on robot learning, meta-learning, and scaling robot learning to broad datasets; her honors include the Sloan Fellowship and the ACM Doctoral Dissertation Award.

Jiaming Song (Luma AI) is Chief Scientist at Luma AI, working on foundation models that reason across modalities and generate rich worlds, spanning architectures, training infrastructure, and data.

Danqi Chen (Princeton) is an Associate Professor at Princeton University who co-leads the Princeton NLP Group, working on natural language processing and large language models.

Stefano Ermon (Stanford) is an Associate Professor at Stanford University working on machine learning and generative AI, with an emphasis on principled methods for real-world problems.

[Zhuang Liu \(Princeton\)](#) is an Assistant Professor at Princeton University working on deep learning and computer vision, and led the development of DenseNet (CVPR Best Paper) and ConvNeXt.

[Zhiqiang Shen \(MBZUAI\)](#) is an assistant professor leading VILA Lab. He focuses on efficient deep learning, machine learning, foundation models, computer vision, and large language models.

[Zico Kolter \(CMU & OpenAI\)](#) is Professor and Head of the Machine Learning Department at CMU and serves on the board of OpenAI, working on the robustness and safety of deep learning.

[Arash Vahdat \(NVIDIA\)](#) is a Research Director leading the generative AI research team at NVIDIA Research, working on generative learning, multimodal training, and generative AI for science.

4 Diversity Commitment

We aim to ensure balance and equity across various dimensions, including backgrounds for invited speakers and organizers, covered topics, and targeted audiences.

Diversity of Organizing Committee. Our organizing committee includes both *male and female* organizers and spans *seven institutions across six regions*: Hong Kong, mainland China, Singapore, South Korea, United Arab Emirates, and United Kingdom. It spans *career stages*, from Ph.D. students to associate professors, and brings together the data-centric learning, foundation-model, and computer vision communities for a cross-disciplinary grounding in the data foundations of modern AI.

Diversity of Speakers. Our invited speakers include both *male and female* researchers and balance *academia and industry*, pairing leading universities (Stanford, Princeton, and CMU) with major industry labs (Luma AI, NVIDIA, and Meta). They span a wide range of *seniority*, from assistant and associate professors to industry research directors and chief scientists, and cover *complementary topics* including robot learning, generative models, vision, language models, and robustness and safety. These areas are all closely tied to data, so the lineup brings diverse yet complementary perspectives that converge on the workshop’s data-centric themes.

Diversity of Participants. We are committed to a diverse and inclusive group of participants. Our outreach will deliberately span disciplines, professional backgrounds, and geographic regions, with particular emphasis on broadening participation from underrepresented communities, so that a wide range of perspectives can engage in and shape the discussion.

5 Anticipated Audience Size and Location Preference

Anticipated audience size. Given the surging interest in data-centric AI, synthetic data, and foundation-model data pipelines, we anticipate approximately 450–500 attendees, comprising both in-person and remote participants.

Location Preference. Any. We have no preference among the candidate locations and are happy to hold the workshop at whichever venue best suits the conference schedule.

6 Difference from Previous Related Workshops

Data has drawn growing attention at recent venues, and some notable workshops have focused on data-centric learning. The broad data-centric lineage began with the Data-Centric AI (DCAI) workshop at NeurIPS 2021, then moved to ICML/ICLR: DataPerf (ICML 2022), followed by the Data-centric Machine Learning Research (DMLR) series (ICML 2023, ICLR 2024, ICML 2024). Other recent venues each address only a *single facet*: ATTRIB (NeurIPS 2023, 2024) for attribution; SyntheticData4ML (NeurIPS 2022, 2023) and AI in the Synthetic Data Age (NeurIPS 2024, 2025) for synthetic data; and DDCV (CVPR 2024) together with the ECCV 2024 Dataset Distillation Challenge for distillation.

Against this fragmented landscape, our workshop is distinguished by five things: *xxx, xxx, xxx, xxx* **To the best of our knowledge, this is the first NeurIPS workshop to treat the entire data revolution behind modern AI systems as a unified, first-class subject of study.**

7 Organizers and Biographies

Contact: We will set up an email group and Slack channel to ensure prompt responses. The public website will list the final contact email once the workshop contact alias is finalized.

Members of the organizing team:

- **Ping Luo** (Associate Professor, HKU, Hong Kong, China), pluo@cs.hku.hk
- **Xinchao Wang** (Associate Professor, NUS, Singapore), xinchao@nus.edu.sg
- **Sukmin Yun** (Assistant Professor, HYU ERICA, Republic of Korea), sukminyun@hanyang.ac.kr
- **Siheng Chen** (Associate Professor, SJTU, China), sihengc@sjtu.edu.cn
- **Bowen Cheng** (AI Research Scientist, Meta, United States), bcheng9@illinois.edu
- **Xinyi Shang** (Ph.D. student, UCL, United Kingdom), xinyi.shang.23@ucl.ac.uk
- **Jiacheng Cui** (Ph.D. student, MBZUAI, United Arab Emirates), jiacheng.cui@mbzuai.ac.ae
- **Peng Sun** (Ph.D. student, ZJU & Westlake University, China), sunpeng@westlake.edu.cn

7.1 Biographies

Ping Luo is an Associate Professor in the Department of Computer Science at the University of Hong Kong and at the HKU Musketeers Foundation Institute of Data Science. His research spans computer vision, deep learning, and machine learning, with notable contributions including Switchable Normalization. He has served as Area Chair for ICCV, CVPR, ECCV, and AAAI.

Xinchao Wang is a tenured Associate Professor at the National University of Singapore, where he directs the xML Lab. His research spans machine learning, computer vision, and deep learning. His honors include the Singapore Young Scientist Award (2025), selection for IEEE “AI’s 10 to Watch” (2025), and the NUS Young Researcher Award (2024);

Sukmin Yun is an Assistant Professor in the Department of Artificial Intelligence at Hanyang University ERICA. His research focuses on efficient AI and model deployment, vision-based and multimodal learning, and foundation models for dynamic environments. His honors include the Qualcomm Innovation Fellowship Korea (2022).

Siheng Chen is an Associate Professor at Shanghai Jiao Tong University. His research covers graph signal processing, graph neural networks, multi-agent collaboration, and autonomous-driving perception. He received the 2018 IEEE Signal Processing Society Young Author Best Paper Award and was selected for China’s National Young Talent Program in 2019.

Bowen Cheng is an AI Research Scientist at Meta. Previously, he worked at OpenAI, and before that, he was a Senior ML Scientist on the Autopilot team at Tesla. He received his Ph.D. in Electrical and Computer Engineering (ECE) from the University of Illinois Urbana-Champaign (UIUC). His research focuses on computer vision and machine learning.

Xinyi Shang is a third-year Ph.D. student in the Department of Statistical Science, University College London. Her research interests are data-centric machine learning, particularly dataset distillation, as well as trustworthy machine learning, such as federated learning and long-tail learning.

Jiacheng Cui is a PhD student at MBZUAI. He received his B.Sc. (Hons) in Artificial Intelligence and Computer Science with First Class Honours from the University of Edinburgh in 2024. He focuses on data-centric AI, dataset distillation, and efficient training.

Peng Sun is a Ph.D. student at Westlake University and Zhejiang University. His research focuses on data-centric learning, generalization, and optimization.

8 Organizing Experience and Team Qualifications

8.1 Previous Organizing Experience

Our team members have extensive experience in organizing workshops and symposiums, making us well-prepared to handle workshop organization at large conferences, with important lists as below:

- Low-Level Vision Frontiers at CVPR 2026 (20 accepted papers) and ECCV 2026;
- Multi-Agent Embodied Intelligent Systems Meet the Generative-AI Era at CVPR 2025 (7 accepted papers), with the End-to-End Autonomous Driving through V2X Cooperation Challenge (30+ teams) and the RoboTwin Dual-Arm Collaboration Challenge (64 teams);
- Multi-Agent Systems in the Era of Foundation Models at ICML 2025;
- Embodied World Models for Decision Making at NeurIPS 2025;
- Cooperative Intelligence for Embodied AI at ECCV 2024 (20 accepted papers);
- Dataset Distillation for Computer Vision at CVPR 2024;
- Special Session on Graph Neural Networks at IEEE ICASSP 2020;
- Statistical Deep Learning in Computer Vision at ICCV 2019 (16 accepted papers);
- Computer Vision for Fashion, Art and Design at ICCV 2019;
- WIDER Face and Pedestrian Challenge at ECCV 2018.

8.2 Team Qualifications

Our organizing team combines senior researchers with extensive experience running international academic events and Ph.D. students serving as motivated first-time organizers, ensuring both operational reliability and fresh energy.

- Prof. Ping Luo serves as Area Chair for CVPR, NeurIPS, ICLR, ICCV, and ICML, including **Notable Area Chair at NeurIPS 2023**. He has co-authored widely read surveys, including *A Survey of Reasoning with Foundation Models: Concepts, Methodologies, and Outlook* (ACM Computing Surveys 2025) and *Autoregressive Models in Vision: A Survey* (TMLR 2025).
- Prof. Xinchao Wang is an Associate Editor of IEEE Transactions on Image Processing (since 2023), IEEE TCSVT, and JVCi; an Area Chair for ECCV 2024 and other CVPR/ICCV/NeurIPS editions; and a Senior Program Committee member for AAAI and IJCAI. He co-authored *A Comprehensive Survey of Dataset Distillation* (IEEE TPAMI 2024), squarely on this workshop’s data-centric theme.
- Prof. Siheng Chen serves as Associate Editor of IEEE Transactions on Signal and Information Processing over Networks and on the NeurIPS 2025 program committee, and authored surveys including *3D Point Cloud Processing and Learning for Autonomous Driving* (IEEE Signal Processing Magazine 2021).
- Prof. Sukmin Yun contributes expertise in efficient and multimodal learning and serves as a reviewer for ICML, NeurIPS, ICLR, CVPR, and AAAI.
- Bowen Chen and the Ph.D. student organizers (Xinyi Shang, Jiacheng Cui, Peng Sun) are first-time workshop organizers active in data-centric research; Xinyi Shang (lead) and Peng Sun co-authored the survey *Dataset Distillation in the Era of Large-Scale Data: Methods, Analysis, and Future Directions* (2025). They will take primary responsibility for day-to-day operations, the reviewing pipeline, and website and communication management.

The organizers’ collective expertise in data-centric learning, efficient learning, and foundation models positions them uniquely to design a program that promotes meaningful discussion on the future of data-centric AI.

9 Program Committee Members

We have secured 50 distinguished members to form our reviewing committee and 10 senior researchers or faculty members to serve as workshop area chairs. For reviewers, approximately 80% of the committee members are senior PhD students and researchers, while the remaining 20% are junior PhD students and senior master students. All reviewers have strong expertise in data-centric AI systems and come from a broad range of leading universities, academic institutions, and industry labs. We are committed to leveraging a global and diverse perspective to ensure a professional, rigorous, and fair review process. Committee recruitment is still ongoing, as we continue to seek additional experts from relevant areas to further strengthen the committee’s expertise and broaden its coverage.

References